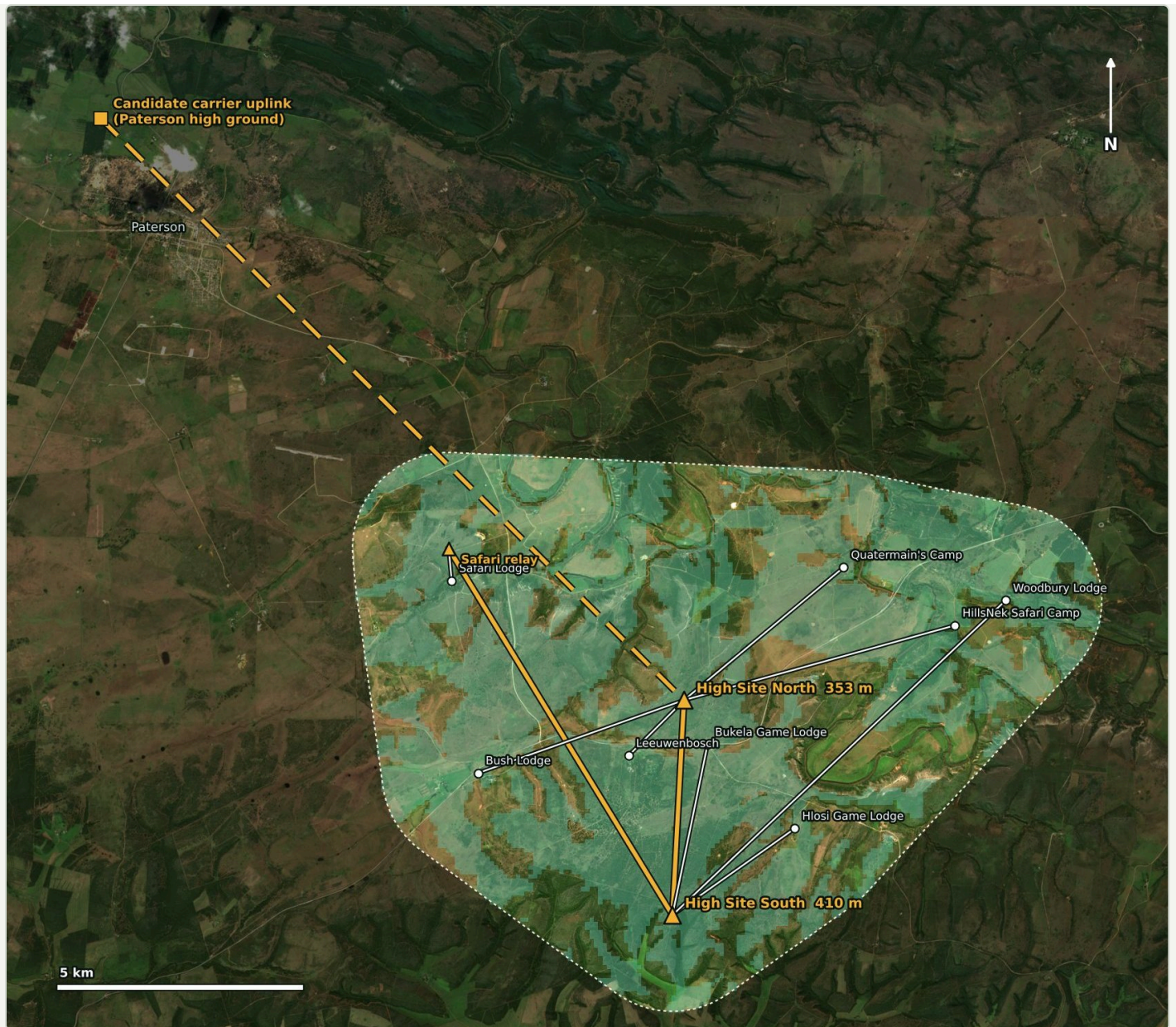


PREPARED FOR RICHARD, AMAKHALA GAME RESERVE

One network across the whole reserve.

We modelled Amakhala's terrain from satellite elevation data and ran every link through a line-of-sight and Fresnel-zone check. The result: **two ridge high-sites and one short relay put 8 of the 8 lodges we could locate on clear radio paths**, fed from a carrier uplink on Paterson high ground.



--- Carrier uplink — Private backbone — Lodge link

Clear line of sight for a 3 m field terminal

Sentinel-2 imagery, 7 September 2026. The dotted outline is the study area around the lodges; everything outside it is dimmed. Site positions are desktop candidates, to be confirmed on a site visit.

8 of 8

lodges on a clear, Fresnel-zone-clear radio path

2 + 1

ridge high-sites plus short relays for lodges in valleys

66%

of the 129 km² study area in line of sight for field teams and sensors

17.1 km

clear uplink path from Paterson high ground into the reserve

HOW IT IS BUILT

Carrier in, private network out

Most lodges on Amakhala list Wi-Fi in the public areas by the main lodge, with each lodge sorting out its own connection. We propose a single backbone that belongs to the reserve.

LAYER 1 • CARRIER

Uplink from Paterson high ground

An uncontended business-grade Vodacom service, delivered by CTTX over a 17.1 km link from Paterson high ground to High Site North. LTE failover keeps the reserve connected if the carrier path drops.

LAYER 2 • BACKBONE

Two ridge high-sites

High Site North (353 m) and High Site South (410 m) sit 4.5 km apart with a clear link between them. A short Safari relay serves a lodge the ridges cannot see. Each runs on solar and lithium storage with remote monitoring, so the network stays up through load-shedding.

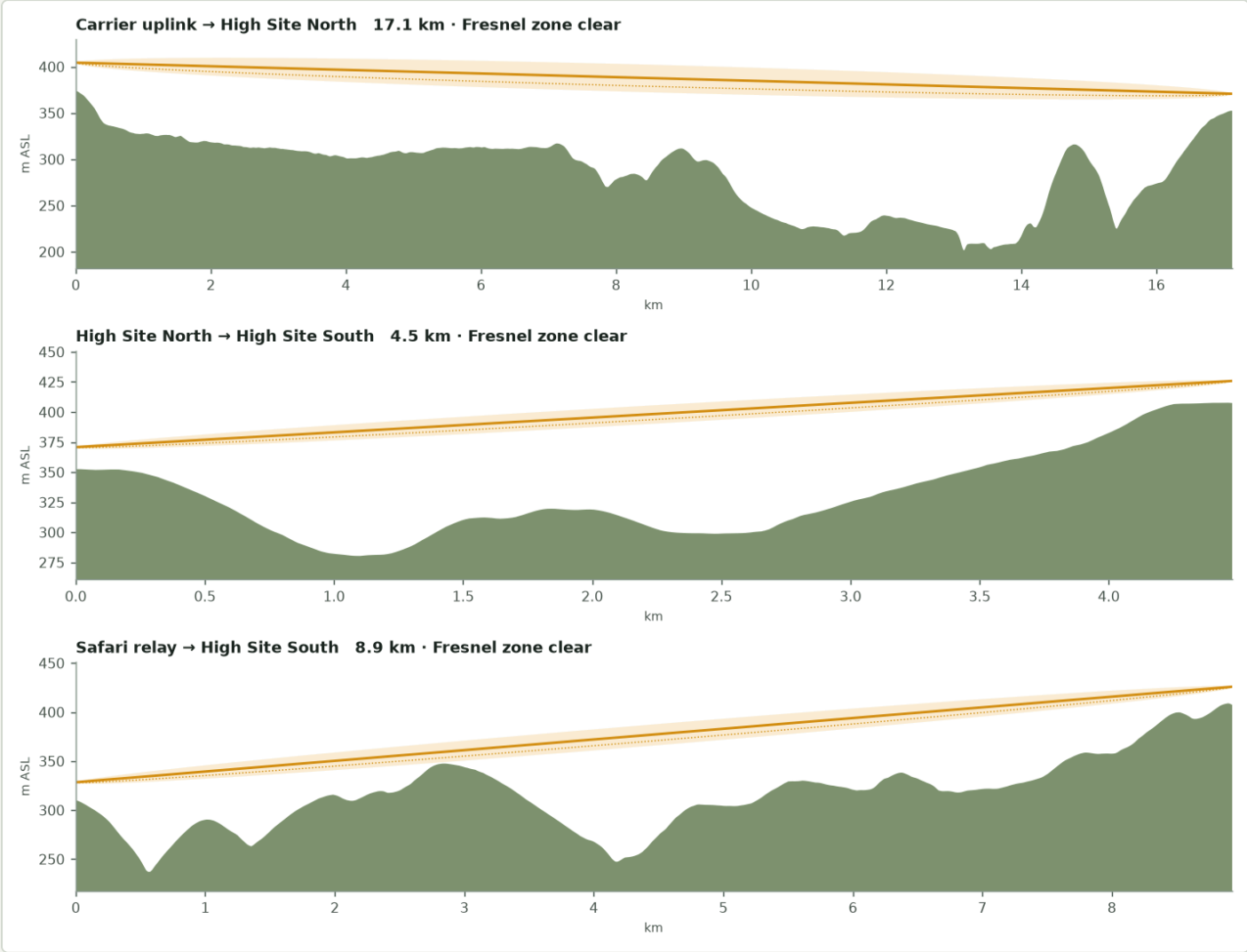
LAYER 3 • DISTRIBUTION

Every lodge and ops site

Sector radios on each high site reach the lodges, HQ, the anti-poaching base, ecology, the workshop and the gates. A LoRaWAN gateway on the backbone is ready for fence, water-point, pump and fuel sensors.

Every path checked against the terrain

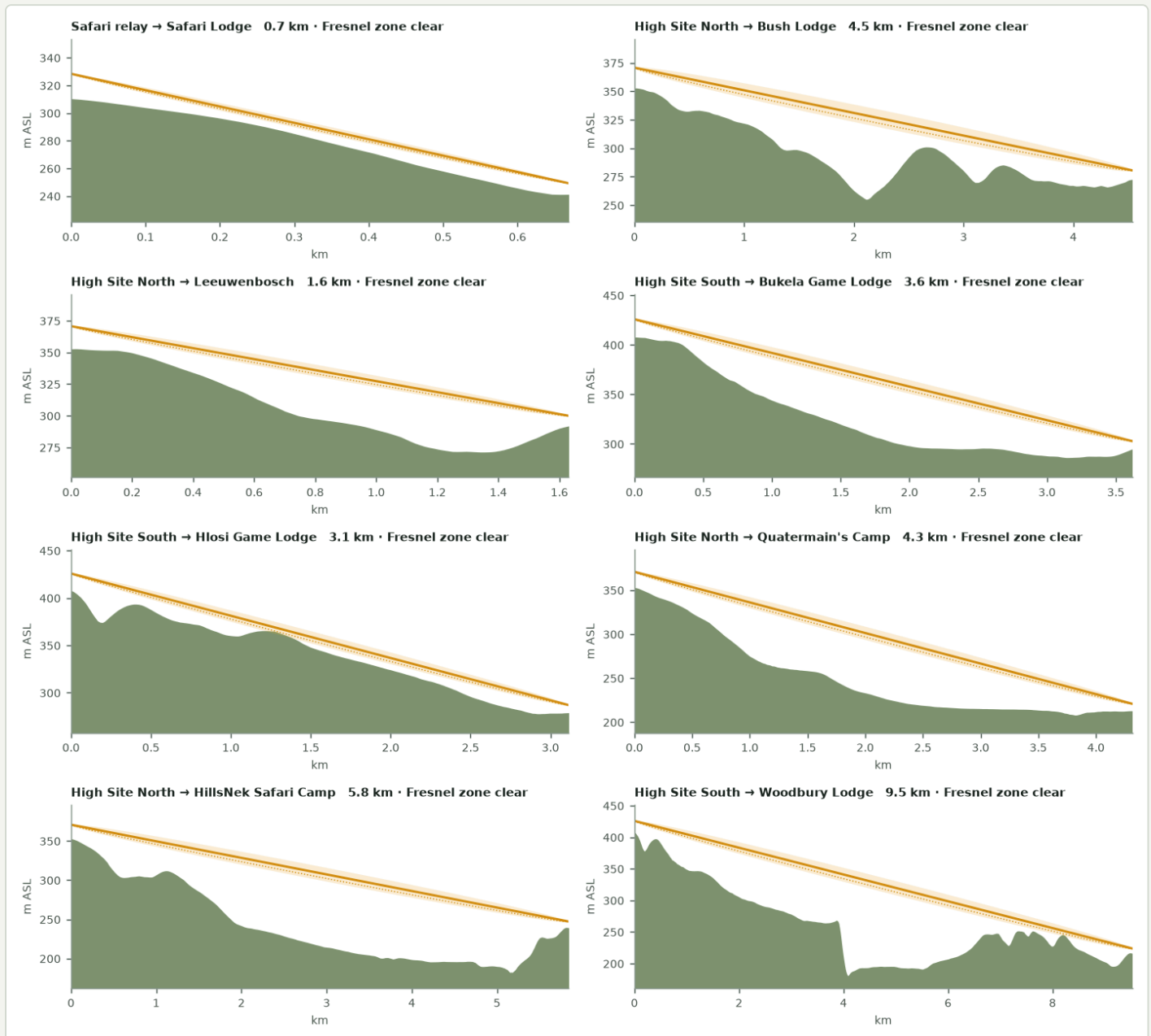
Each profile below shows the ground between two sites (green), the radio path (amber line) and its first Fresnel zone (amber band). We allow for earth curvature and require at least 60% of the Fresnel zone to be clear at every point. A path that fails is not drawn on the map.



Backbone and uplink paths. 5.8 GHz, $k = 4/3$, 30 m uplink tower, 18 m high-site masts.

LODGE	SERVED FROM	PATH LENGTH	LINE OF SIGHT
Safari Lodge	Safari relay	0.7 km	Clear
Bush Lodge	High Site North	4.5 km	Clear
Leeuwenbosch	High Site North	1.6 km	Clear
Bukela Game Lodge	High Site South	3.6 km	Clear

LODGE	SERVED FROM	PATH LENGTH	LINE OF SIGHT
Hlosi Game Lodge	High Site South	3.1 km	Clear
Quatermain's Camp	High Site North	4.3 km	Clear
HillsNek Safari Camp	High Site North	5.8 km	Clear
Woodbury Lodge	High Site South	9.5 km	Clear



Lodge paths, with an 8 m receiver mast at each lodge. Lodge positions come from published GPS coordinates.

One backbone for every driver of the business

Carrier connectivity comes in once. Everything on the property runs over infrastructure the reserve owns. New cameras, sensors and lodges plug in without a new provider contract.

Guest experience

Reliable Wi-Fi, booking and card payments at every lodge.

Staff communication

Lodges, HQ, rangers and maintenance on one network, day and night.

Security and operations

Cameras, gates, fence alarms and sensors on the same backbone, powered through load-shedding.

Sandymount Safaris · 30,000+ ha, three lodges

BEFORE

A consumer wireless service at about 3 Mbps down and 8 Mbps up, with the provider on site daily to swap IPs and reinstall equipment.

AFTER

A Vodacom-backed 100 Mbps symmetrical link into the reserve, a CTTX-built main mast and two secondary masts, solar and lithium power, all monitored remotely.

GUEST EXPERIENCE

HD sport streams live at the lodges. Guest connectivity is no longer something the lodges think about.

STAFF COMMUNICATION

Over 800 Mbps measured across the reserve's own backbone, linking the main lodge, Mountain Retreat and Samar.

SECURITY AND OPERATIONS

Daily provider callouts went to zero. Every fault used to need a vehicle and a person; now the network is monitored remotely.

OWNERSHIP

The reserve keeps adding systems on top of the backbone. In the client's words, they stopped doing speed tests.

Elephants broke out after 21:00 and staff had no effective way to coordinate. The requirement that came out of it: dependable communications after hours, across the areas where an incident can happen.

An Eastern Cape lodge near Addo

What changes when the reserve owns its network

Today: a connection per lodge

- Each lodge buys and manages its own service
- Wi-Fi is listed for the public areas by the main lodge
- Security, ecology and maintenance teams are not on a shared network
- Nothing links the gates, water points or fences

With a reserve backbone

- One uncontended carrier feed shared across the reserve
- Lodges, HQ, the anti-poaching base and gates on one private network
- Coverage out into the field, day and night
- An owned asset that takes cameras and sensors without new towers

Avoided loss

Gate cameras, fence and trigger alarms, and after-hours communication for the Equine APU and K9 unit. When something happens at night, the right people know in minutes.

Infrastructure ownership

High sites, backbone and power belong to the reserve. The network grows with Amakhala instead of depending on each lodge's separate provider.

Operational efficiency

Water points, pumps and fuel tanks report their own status, so there are fewer vehicle trips just to check. Maintenance, ecology and lodges share one communications path.

Executive risk reduction

Management gets live visibility of the network, cameras and sensors, plus a record of operational diligence for owners, insurers and conservation partners.

NEXT STEP

Check it against what you see on the ground

30-minute call

Your peak usage, the incidents hardest to coordinate, and what currently needs a vehicle trip to check.

Confirm the sites

A marked map of HQ, the anti-poaching base and the gates. We add them to the model and rerun the link plan.

One day on the reserve

We walk the high sites and relay points and confirm the carrier uplink with Vodacom.

Design and proposal

A final network design in Cambium LINKPlanner, with equipment, power and a phased plan.

Method. Terrain: Copernicus DEM GLO-30 (30 m). Imagery: ESA Sentinel-2 L2A true colour, 5 August 2026. Radio model: 5.8 GHz, effective earth radius factor $4/3$, 60% first-Fresnel-zone clearance required at every point. Heights: carrier uplink 30 m, high-site masts 18 m, lodge masts 8 m, field terminals 3 m. The DEM does not model trees or buildings. Final paths are confirmed on survey and in Cambium LINKPlanner before anything is built.

Candidate positions. High Site South -33.56661, 26.07792; High Site North -33.52632, 26.08093; Safari relay -33.49751, 26.02860; carrier uplink -33.41625, 25.95175 (Paterson high ground). The exact carrier site is subject to Vodacom feasibility.